1D Ring Two-Walker Encounter With Shortcut

Robust A1/A8 Verification and Bimodal Realization

valley-k-small report extension

March 21, 2026

Abstract

This report delivers the requested 1D ring encounter extension in two blocks: (1) an implementation-ready and numerically robust verification of Eq. (A1)⇔Eq. (A8) from the appendix "Product of two generating functions"; (2) an exact two-walker encounter solver on a 1D ring with a directed shortcut, including a shortcut-channel decomposition that explains the observed double-peak regime. In the present report, the production solver for the main encounter curves is the exact time-domain pair recursion; the A1/A8 block is the transform-domain compatibility anchor. For a family-level runtime comparison between the Luca / generating-function route and the time-domain route, see the unified benchmark companion report `luca_vs_recursion_unified_benchmark`. All figures, tables, and machine-readable outputs are generated from one script.

1 Problem Setup

We study two independent lazy biased walkers on the same periodic 1D ring. Encounter first-passage is defined as

$$T_{\text{enc}} = \min\{t \geq 1 : X_t = Y_t\}, \qquad f_{\text{enc}}(t) = \mathbb{P}(T_{\text{enc}} = t).$$

Walker A has a directed shortcut src → dst with strength β; walker B has no shortcut.

Default representative parameters:

$$N = 101,\ q = 0.70,\ g_1 = 0.70,\ g_2 = -0.40,\ n_0 = 5,\ m_0 = 55,\ \text{src} = 5,\ \text{dst} = 70.$$

Directed Shortcut Injection (Selfloop Mode)

Walker A uses the same shortcut convention as the ring shortcut reports:

$$p_{\text{sc}} = \beta(1-q), \qquad 0 \leq p_{\text{sc}} \leq 1-q.$$

At node src we transfer this mass from self-loop to the directed edge:

$$P_1(\text{src}, \text{src}) = (1-q) - p_{\text{sc}},$$

$$P_1(\text{src}, \text{dst}) = P_1^{(0)}(\text{src}, \text{dst}) + p_{\text{sc}},$$

while all other entries follow the baseline lazy-biased kernel $P_1^{(0)}$. This keeps each row stochastic and preserves the interpretation of β as shortcut strength.

Unified Notation and Metrics

To make this first dedicated encounter report unambiguous, we fix all quantities used below:

- One-step kernels: P_1, P_2; non-absorbed joint law $J_t(i,j) = \Pr(X_t = i, Y_t = j, T_{\text{enc}} > t)$.
- Two encounter times:

$$T_{\text{any}} = \min\{t \geq 1 : X_t = Y_t\}, \quad T_\delta = \min\{t \geq 1 : X_t = Y_t = \delta\}.$$

- First-passage pmf and survival:

$$f(t) = \Pr(T = t), \quad S(t) = \Pr(T > t) = 1 - \sum_{\tau=1}^{t} f(\tau).$$

- Detector traces used for peak/valley metrics:

$$\bar{f}(t) = \text{MA}_w[f](t),$$

  with fixed odd window $w = 11$ for anywhere-encounter scans; for fixed-site K=2 diagnostics, use

$$\bar{\widetilde{f}}(m) = \text{MA}_9[\widetilde{f}](m).$$

- Peak/valley detector times (under one fixed smoothing+threshold protocol):

  anywhere: t_1 = first prominent local peak in $[t_{\text{ignore}}, t_{\text{end}}]$, $\quad t_2$ = selected comparison peak in $(t_1, t_{\text{end}}]$, $\quad t_v$

  fixed-site K=2: run the same selector on parity index m of $g_m = \bar{\widetilde{f}}(m)$, with $\tau \geq t_{\text{ignore,pair}}$, $\tau - t_1 \geq \text{min_s}$

- Peak/valley intensity ratios (written on the detector trace g: anywhere uses $g = \bar{f}$, fixed-site K=2 uses $g = \bar{\widetilde{f}}$ with the $t = 2m$ map):

$$R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}, \quad R_{\text{peak}} = R_{\text{peak}}(t_2), \qquad R_{\text{dir}} = \frac{g_{t_2}}{g_{t_1}}, \qquad R_{\text{valley}} = \frac{g_{t_v}}{\max\{g_{t_1}, g_{t_2}\}}.$$

  Candidate scoring uses $R_{\text{peak}}(\tau), R_{\text{valley}}(\tau)$ to choose the late partner peak, then commits to $R_{\text{peak}}, R_{\text{valley}}$ at $\tau = t_2$. By definition $R_{\text{dir}} = g_{t_2}/g_{t_1}$ may exceed 1; all three ratios are reported as diagnostics, but the phase/onset gate itself is now based on peak separation rather than ratio thresholds.

- Phase labels: single / weak / clear (encoded as 0/1/2). We report two onset notions in parallel: the first shortcut strength with at least two prominent peaks ('has≥2 onset'), and the first shortcut strength entering clear phase ('clear onset').

- Shortcut-channel shares:

$$h_{\text{sc}}(t) = \frac{f_{\text{sc}}(t)}{f_{\text{sc}}(t) + f_{\text{no}}(t)}, \quad H_{\text{sc}}(T) = \frac{\sum_{t \leq T} f_{\text{sc}}(t)}{\sum_{t \leq T}(f_{\text{sc}}(t) + f_{\text{no}}(t))}.$$

- Mass-conservation audit:

$$\varepsilon_{\text{mass}} = \left| 1 - \sum_{t=1}^{t_{\max}} f(t) - S(t_{\max}) \right|.$$

Mathematical Phase Definition (0/1/2)

Let g denote the detector trace ($g = \bar{f}$ for anywhere; $g = \bar{\widetilde{f}}$ for fixed-site K=2 coarse diagnostics). Define prominent local-peak candidates

$$\mathcal{P} = \{\tau : \; g_{\tau-1} < g_\tau, \; g_\tau \geq g_{\tau+1}, \; g_\tau \geq 0.01 \max g\},$$

plus detector-window constraints:

$$\tau \in [t_{\text{ignore}}, t_{\text{end}}] \text{ (anywhere)}, \qquad \tau \geq t_{\text{ignore,pair}}, \; \tau - t_1 \geq \text{min_sep_pair (fixed-site K=2 coarse)}.$$

If $|\mathcal{P}| < 2$, no valid peak pair is available and we set 'has_two=False'. We keep only candidates with $\tau - t_1 \geq 2$ so $t_v(\tau) = \arg\min_{t_1<t<\tau} g_t$ is defined on a strict interior interval (t_1, τ). Otherwise, we choose (t_1, t_2) by the configured selector:

$$t_1 = \min \mathcal{P}, \qquad t_2 = \arg \max_{\tau \in \mathcal{P}, \; \tau > t_1} \frac{(\tau - t_1)\, R_{\text{peak}}(\tau)}{R_{\text{valley}}(\tau) + 10^{-12}},$$

where the fixed-site K=2 branch applies the additional parity-index constraint $\tau - t_1 \geq$ min_sep_pair and reports times after the $t = 2m$ back-map to the raw clock. If scores tie within numerical tolerance ($|\Delta\text{score}| \leq 10^{-12}$), we break ties by larger τ, then smaller R_{valley}, then larger R_{peak}. with

$$t_v(\tau) = \arg \min_{t_1<t<\tau} g_t, \quad R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}, \quad R_{\text{valley}}(\tau) = \frac{g_{t_v(\tau)}}{\max\{g_{t_1}, g_\tau\}}.$$

After selecting t_2, define

$$R_{\text{peak}} = R_{\text{peak}}(t_2), \qquad R_{\text{valley}} = R_{\text{valley}}(t_2), \qquad \text{sep}_{\text{peaks}} = \frac{|t_2 - t_1|}{w_{1/2}^{(1)} + w_{1/2}^{(2)}}.$$

Here $w_{1/2}^{(1)}, w_{1/2}^{(2)}$ are the half-widths at half-maximum of the selected peaks, measured directly on the detector trace g. The phase code is

$$\text{phase} = \begin{cases} 0, & \neg\text{has_two}, \\ 2, & \text{has_two} \;\wedge\; \text{sep}_{\text{peaks}} \geq 1, \\ 1, & \text{has_two} \;\wedge\; \text{sep}_{\text{peaks}} < 1. \end{cases}$$

Anywhere primary detector settings are $(w, t_{\text{ignore}}, t_{\text{end}}) = (11, 80, 400)$. Fixed-site coarse settings are $(w, t_{\text{ignore,pair}}, \text{min_sep_pair}) = (9, 18, 8)$ in parity-coarsened index. The separation threshold is uniformly $\text{sep}_{\text{peaks}} \geq 1$. The ratio quantities $R_{\text{peak}}, R_{\text{dir}}, R_{\text{valley}}$ remain in the tables and figures as secondary diagnostics, and $R_{\text{peak}}(\tau), R_{\text{valley}}(\tau)$ still appear in the candidate-selection score; they are simply no longer extra phase gates.

2 Compact Derivation Roadmap: A1 to A8

Define

$$\alpha_k = 1 + s_k = 1 + q\left(\cos\frac{2\pi k}{N} - 1\right), \qquad \widetilde{P}_{n_0}(n, u) = \sum_{k=0}^{N-1} \frac{h_k(n, n_0)}{1 - u\alpha_k}.$$

Insert into A1:

$$I(z) = \frac{1}{2\pi i}\oint_C \frac{du}{u}\widetilde{P}_{n_0}(n, u)\widetilde{P}_{m_0}\left(n, \frac{z}{u}\right) = \frac{1}{2\pi i}\oint_C \sum_{k_1,k_2} \frac{h_{k_1}h_{k_2}}{(1 - u\alpha_{k_1})(u - z\alpha_{k_2})} du.$$

For each (k_1, k_2), poles are

$$u = \frac{1}{\alpha_{k_1}} \quad \text{(outer poles)}, \qquad u = z\alpha_{k_2} \quad \text{(inner poles)}.$$

Choose $|u| = r$ such that

$$\max_k |z\alpha_k| < r < \min_{k:\alpha_k \neq 0} \left| \frac{1}{\alpha_k} \right|.$$

Then only inner poles are enclosed, and residue summation gives

$$I(z) = \sum_{k_1,k_2} \frac{h_{k_1}(n, n_0) h_{k_2}(n, m_0)}{1 - z\alpha_{k_1}\alpha_{k_2}},$$

which is exactly Eq. (A8).

Encounter Mapping (fixed-site and anywhere)

The same A1→A8 block is the core kernel for both encounter definitions:

- fixed-site encounter at δ:
$$Q_\delta(t) = P_{n_0}(\delta, t) P_{m_0}(\delta, t),$$
so A1/A8 applies directly at $n = \delta$.

- anywhere encounter:
$$Q_{\text{any}}(t) = \sum_{n=0}^{N-1} P_{n_0}(n, t) P_{m_0}(n, t),$$
hence
$$\widetilde{Q}_{\text{any}}(z) = \sum_{n=0}^{N-1} \frac{1}{2\pi i} \oint_C \frac{du}{u} \widetilde{P}_{n_0}(n, u) \widetilde{P}_{m_0}\left(n, \frac{z}{u}\right).$$

For the homogeneous defect-free ring, translation invariance lets the modal double sum collapse to a single sum:

$$\widetilde{Q}_{\text{any}}(z) = \sum_{k=0}^{N-1} \frac{\frac{1}{N} e^{i2\pi k(m_0 - n_0)/N}}{1 - z\,\lambda_k^{(1)} \lambda_{-k}^{(2)}},$$

where $\lambda_k^{(1)}, \lambda_k^{(2)}$ are one-step mode eigenvalues of walkers A/B. This collapse should therefore be read as a transform-domain validation block for the homogeneous benchmark. In the main shortcut model studied later, walker A carries a local directed defect at src → dst, so translation invariance is broken and we do not rely on the Fourier-collapse formula there. All production anywhere-encounter figures with shortcut are generated instead by exact time-domain absorption recursion.

3 Numerically Stable Verification Routine

Use $u = re^{i\theta}$ and $du/u = i\,d\theta$:

$$I(z) = \frac{1}{2\pi} \int_0^{2\pi} \widetilde{P}_{n_0}(n, re^{i\theta}) \widetilde{P}_{m_0}\left(n, \frac{z}{re^{i\theta}}\right) d\theta \approx \frac{1}{M} \sum_{j=0}^{M-1} F\left(re^{2\pi i j/M}\right).$$

Implementation choices used in code:

1. enforce contour feasibility with $r_{\text{in}} < r < r_{\text{out}}$;

2. choose $r = \sqrt{r_{\text{in}} r_{\text{out}}}$ by default;

3. adaptive doubling in M from 16 upward until $|I_{2M} - I_M| < \varepsilon(1 + |I_{2M}|)$;

4. cross-check contour LHS against direct RHS (A8).

Four sanity cases (real/complex z, edge $q = 0$, small $|z|$) all pass with relative errors around $10^{-16} \sim 10^{-18}$.

Table 1: A1/A8 numerical verification summary.

Case	(N, q)	z	M	r	$\lvert LHS - RHS \rvert$	rel.err
R1_real_small	(16,0.30)	$0.40 + 0.00i$	128	0.632	8.13e-18	8.13e-18
R2_complex	(12,0.35)	$0.30 + 0.10i$	64	0.562	1.52e-17	1.52e-17
R3_edge_q0	(20,0.00)	$0.55 + 0.00i$	256	0.742	4.44e-16	1.38e-16
R4_small_z	(18,0.40)	$0.05 + 0.20i$	64	0.454	4.53e-18	4.53e-18

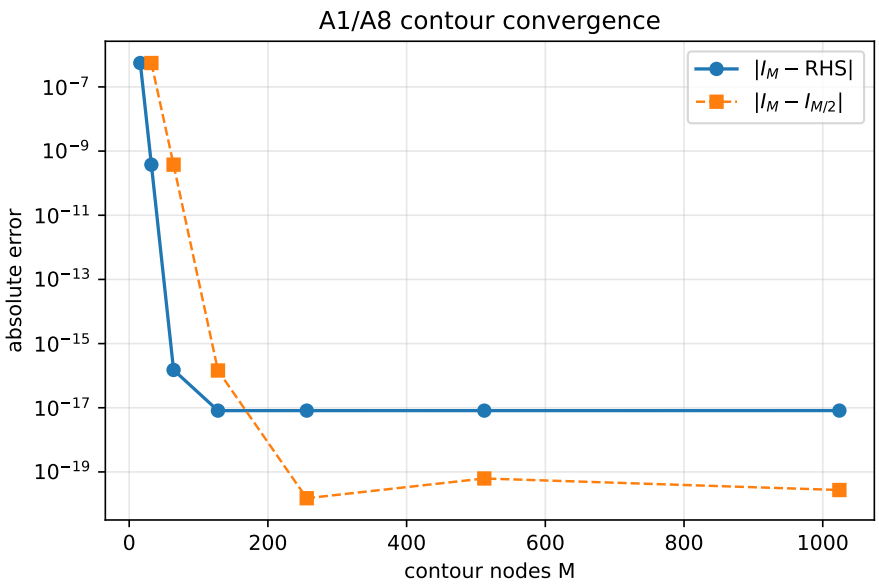

Figure 1: Error decay versus contour node count M.

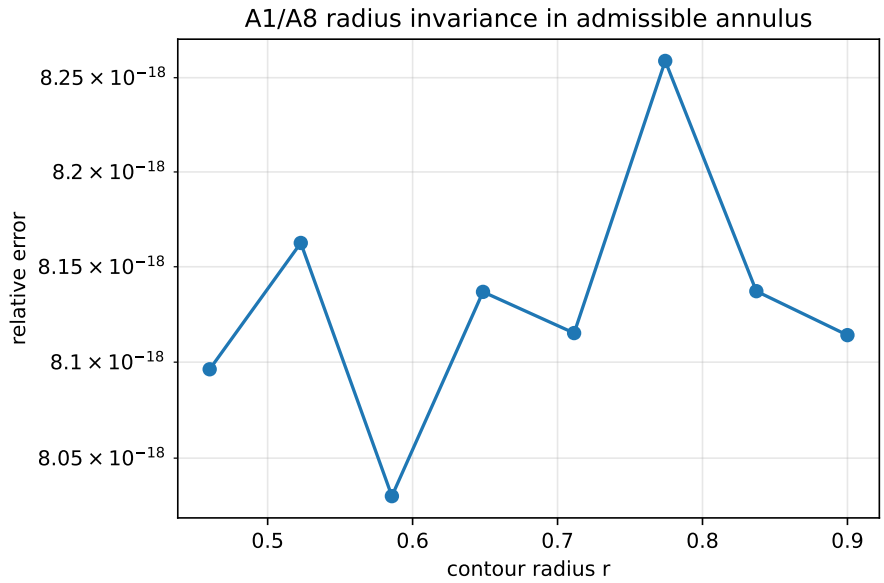

Figure 2: Radius invariance inside admissible annulus.

4 1D Ring Encounter Algorithm

Let

$$J_t(i,j) = \mathbb{P}(X_t = i, Y_t = j, T_{\text{enc}} > t).$$

Exact update per time step:

$$J_{t+1} = P_1^\top J_t P_2, \qquad f_{\text{enc}}(t+1) = \text{tr}(J_{t+1}).$$

After extracting encounter flux, the diagonal of J_{t+1} is set to zero (absorbing condition). This gives deterministic, noise-free FPT curves.

A second coupled update tracks two channels:

- trajectories that have used the shortcut at least once;
- trajectories that have never used the shortcut.

Their sum equals the total encounter FPT and explains peak origins.

Mathematical Mechanism of Bimodality

In this 1D ring encounter model, the double-peak regime is produced by superposition of two channel-specific time scales rather than numerical artifacts:

$$f_{\text{enc}}(t) = f_{\text{sc}}(t) + f_{\text{no}}(t).$$

1. **Early lobe:** shortcut-used trajectories dominate around the first peak (high $h_{\text{sc}}(t)$), corresponding to shorter effective paths.
2. **Late lobe:** no-shortcut trajectories take over in the mid/late window and generate the second peak scale.
3. **Ring-topology amplification:** relative displacement and winding on a periodic ring increase delay/spread of the late channel, enlarging $t_2 - t_1$.

For the representative case (Table 5), we observe $t_1 < t_v < t_2$ with threshold-qualified $R_{\text{peak}}, R_{\text{valley}}$; R_{dir} is reported only as a directional diagnostic (not used in phase thresholds). Together with the channel handoff in the shortcut-share curve, this closes the mechanism chain for a clear-bimodal diagnosis. The important nuance is that a weak double / multi-timescale structure is already visible at $\beta = 0$; the shortcut does not create bimodality ex nihilo, but lifts an earlier fast channel above threshold and reorganizes the signal into a cleaner two/three-peak regime.

5 Fixed-Site Encounter Extension ($n = m = \delta$)

We also complete the fixed-site encounter branch highlighted in your notes:

$$T_\delta = \min\{t \geq 1 : X_t = Y_t = \delta\}.$$

So we track two encounter definitions in one framework:

- anywhere encounter: $T_{\text{any}} = \min\{t \geq 1 : X_t = Y_t\}$;
- fixed-site encounter: $T_\delta = \min\{t \geq 1 : X_t = Y_t = \delta\}$.

For fixed-site, define the free-process synchronous occupancy

$$Q_{n_0,m_0}(t) = \Pr(X_t = \delta, Y_t = \delta \mid X_0 = n_0, Y_0 = m_0) = P^{(1)}_{n_0}(\delta, t)\, P^{(2)}_{m_0}(\delta, t).$$

Its generating function is exactly the appendix product-of-generating-functions object, evaluable by A1 contour integration or A8 modal double sum. Then renewal gives

$$\widetilde{f}_\delta(z) = \frac{\widetilde{Q}_{n_0,m_0}(z)}{\widetilde{Q}_{\delta,\delta}(z)}.$$

Numerically, we implement exact time-domain absorption at a single point (δ, δ) and apply K=2 parity coarse graining:

$$\widetilde{f}(0) = f(0), \qquad \widetilde{f}(m) = f(2m-1) + f(2m), \quad m \geq 1.$$

For first-passage traces, $f(0) = 0$ so diagnostics effectively start from $m \geq 1$; reported detector times use the right-endpoint map $t = 2m$.

At baseline $N = 30, q = 0.7, n_0 = 5, m_0 = 12, \delta = 0$, drift cases show:

- $(g_1, g_2) = (0, 0)$ is unimodal;
- $(0.9, 0.9)$, $(0.7, 0.7)$, and $(0.9, -0.9)$ are clearly bimodal.

The full phase-map occupancy (single/weak/clear fractions over the full drift grid) is auto-summarized in Table 2. This fixed-site branch is intentionally a drift-dominated benchmark, so the resulting clear-phase occupancy should be read as benchmark-specific rather than a universal statement about all fixed-site encounter settings.

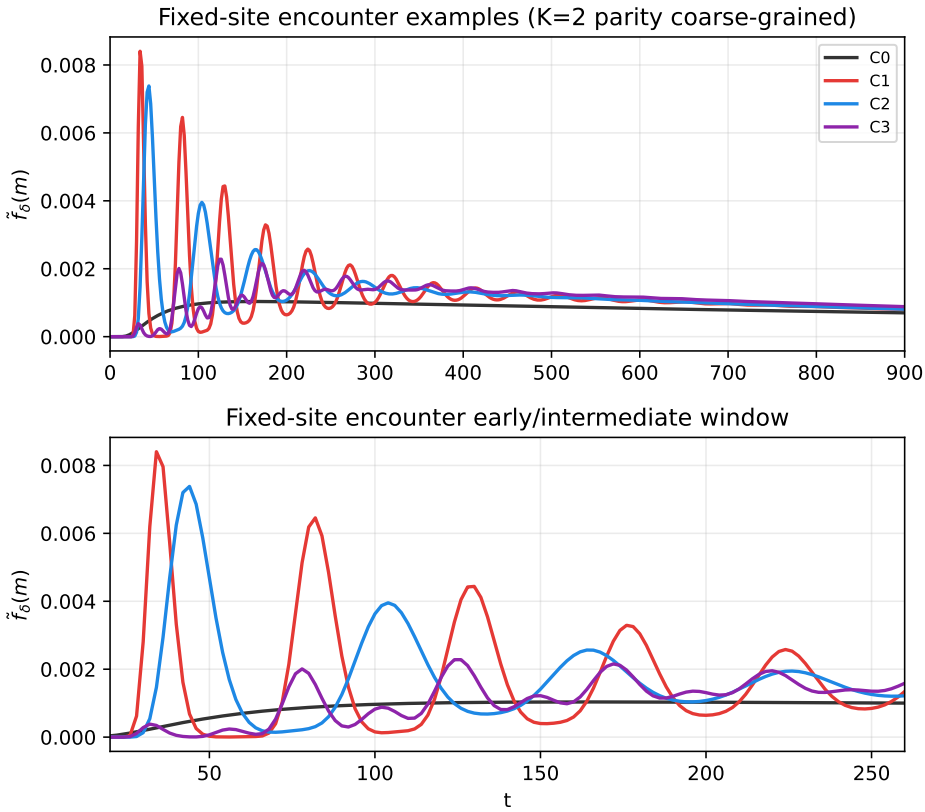

Figure 3: Fixed-site encounter examples after K=2 parity coarse graining.

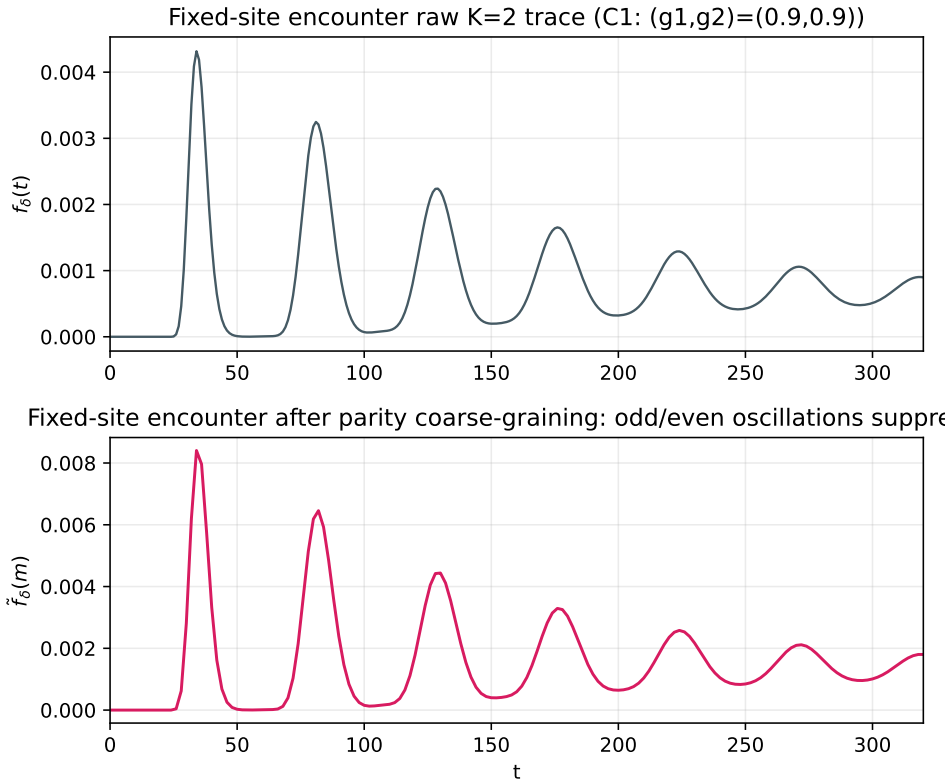

Figure 4: Why fixed-site encounter parity coarse graining is needed for K=2: raw trace (odd-even micro-oscillation) versus parity-coarsened trace used by the detector.

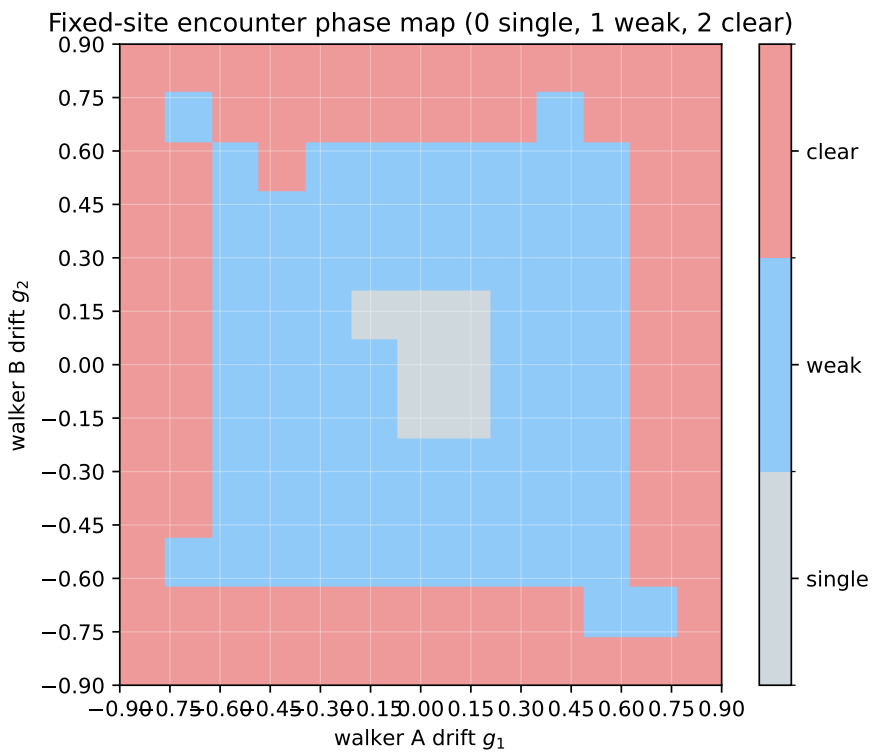

Figure 5: Fixed-site phase map on (g_1, g_2): 0 single, 1 weak, 2 clear.

Table 2: Fixed-site phase-map occupancy summary on the (g_1, g_2) grid.

Phase class	Count	Fraction
single (0)	7	0.041
weak (1)	78	0.462
clear (2)	84	0.497

The clear fraction is $84/169 = 0.497$ (auto-synced from generated scan data). This fixed-site benchmark is intentionally drift-dominated; K=2 parity coarse graining suppresses odd-even micro-structure so classification tracks separated time scales rather than parity artifacts. For first-passage traces, $f(0) = 0$, so $\widetilde{f}(0) = f(0) = 0$ and diagnostics effectively start at $m \geq 1$. As in the main text, candidate peak pairs require $\tau - t_1 \geq 2$ so $t_v(\tau) = \arg\min_{t_1<t<\tau} g_t$ is defined on a strict interior interval. This branch uses the same timescale selector as the main scan: detector trace $g_m = \bar{\widetilde{f}}(m)$, first peak + score-selected t_2 ($R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}$, $R_{\text{peak}} = \min / \max$ (i.e., $R_{\text{peak}}(t_2)$), $R_{\text{dir}} = g_{t_2}/g_{t_1}$ as a directional diagnostic only (it may exceed 1 and does not enter phase thresholds), $\text{score}(\tau) = \frac{(\tau - t_1)\, R_{\text{peak}}(\tau)}{R_{\text{valley}}(\tau) + 10^{-12}}$). If $|\Delta\text{score}| \leq 10^{-12}$, tie-break order is larger τ →smaller valley ratio→larger peak balance. With parity-index constraint $\tau - t_1 \geq$ min_sep_pair, no extra t_{end} cutoff, and $t = 2m$ back-mapping.

Table 3: Fixed-site example metrics.

Case	(g_1, g_2)	phase	$t_1\,(= 2m)$	$t_2\,(= 2m)$	sep ($\bar{\widetilde{f}}$)	peak balance ratio ($\bar{\widetilde{f}}$, min/max)	valley rati
C0	(0.00,0.00)	0	-	-	–	–	–
C1	(0.90,0.90)	2	36	740	1.751	0.225	0.003
C2	(0.70,0.70)	2	44	518	1.137	0.210	0.035
C3	(0.90,-0.90)	2	78	592	1.251	0.864	0.424

In this table, t_1, t_2 are reported after mapping parity-coarsened indices back to the raw clock by $t = 2m$; both peak-balance and valley ratios are computed on the smoothed coarse trace $\bar{\widetilde{f}}(m)$, so parity micro-oscillations do not enter fixed-site phase labels. When no second peak is detected (t_1, t_2 absent), both ratios are reported as "–" to mark the values as undefined rather than numeric 0/1.

6 Bimodality Results for 1D Ring Encounter

Scanning $\beta \in [0, 0.30]$ uses a fixed diagnostic protocol (same smoothing and $t_{\text{ignore}} = 80$). Detector-perturbation settings are fixed as smooth $\in \{9, 11, 13\}$, $t_{\text{ignore}} \in \{60, 80, 100\}$, min-ratio $\in \{0.18, 0.20, 0.22\}$, valley-cap $\in \{0.85, 0.90, 0.95\}$. Unless explicitly marked as a legacy comparison, onset/phase/representative metrics in this report use the timescale detector below as the primary selector.

Multi-Peak Selection Rule (implementation-level)

To avoid ambiguity when three or more prominent peaks coexist, we use the following deterministic rule: Let g denote the detector trace (anywhere: $g = \bar{f}$; fixed-site K=2: $g = \bar{\widetilde{f}}$ with $t = 2m$ mapping).

1. Anywhere scan: smooth the raw first-passage sequence into g with the fixed odd window, and keep candidate peaks in $[t_{\text{ignore}}, t_{\text{end}}]$.

2. Fixed-site K=2 coarse branch: run the same score-based selector on parity index m, with candidates constrained by $\tau \geq t_{\text{ignore,pair}}$ and $\tau - t_1 \geq$ min_sep_pair; this branch does not apply an extra t_{end} cutoff, and reported times are mapped back by $t = 2m$.

3. Set t_1 as the earliest prominent peak passing amplitude threshold.

4. For each later candidate $\tau > t_1$, define (candidate pairs are restricted to $\tau - t_1 \geq 2$ to keep the valley definition interior)

$$R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}, \quad t_v(\tau) = \arg\min_{t_1 < t < \tau} g_t, \quad R_{\text{valley}}(\tau) = \frac{g_{t_v(\tau)}}{\max\{g_{t_1}, g_\tau\}},$$

$$\text{score}(\tau) = \frac{(\tau - t_1) \cdot R_{\text{peak}}(\tau)}{R_{\text{valley}}(\tau) + 10^{-12}}.$$

5. Choose $t_2 = \arg\max_{\tau > t_1} \text{score}(\tau)$; if scores tie up to numerical tolerance ($|\Delta\text{score}| \leq 10^{-12}$), choose the larger τ (later timescale); if still tied, choose smaller valley ratio; if still tied, choose larger peak-balance ratio.

This "timescale detector" keeps the late-timescale peak stable in multi-peak windows and prevents non-physical t_2 switching.

Key numeric claims are inserted directly from generated outputs:

1. No resolvable phase-1 plateau appears inside the current scan window; once two peaks are detected, the clear/phase-2 onset nearly coincides with the has≥2 onset.

2. coarse scan first reaches has≥2 prominent peaks at $\beta \approx 0.00$, and enters phase 2 once $\text{sep}_{\text{peaks}} \geq 1$ at $\beta \approx 0.00$;

3. representative case $\beta = 0.20$: peaks at $t_1 = 108$, $t_2 = 321$, separation $\text{sep}_{\text{peaks}} = 2.766$, peak-balance ratio $R_{\text{peak}} = 0.306$ (min/max), directed peak ratio $R_{\text{dir}} = 0.306$ (diagnostic only; may exceed 1; not used in phase thresholds), valley ratio $R_{\text{valley}} = 0.126$.

4. A weak double / multi-timescale structure already exists at $\beta = 0$; the shortcut does not create the phenomenon ex nihilo, but lifts an earlier fast channel above threshold and reorganizes the overall signal into a cleaner two/three-peak regime.

We further refine onset on $\beta \in [0.00, 0.05]$ with step 0.005, giving nominal has≥2 onset $\beta \approx 0.00$ and nominal clear onset $\beta \approx 0.00$. Under detector perturbations, has≥2 onset range is $[0.00, 0.00]$ with median 0.00; clear onset range is $[0.00, 0.00]$ with median 0.00; detector-agreement first crosses 50% at $\beta \approx 0.00$ and 75% at $\beta \approx 0.00$, with agreement width $\Delta\beta_{25\to75} \approx 0.00$. Mass-conservation check: $\sum_t f(t) = 0.99999994$, $S(t_{\max}) \approx 5.88e-08$.

To keep iterative runs synchronized with narrative claims, key numeric indicators are auto-generated from machine outputs:

Table 4: Auto-generated key metrics for the 1D ring encounter scan.

Metric	Value
Has≥2 onset β (coarse/refined)	0.000 / 0.000
Clear onset β (coarse/refined)	0.000 / 0.000
Has≥2 onset range	[0.000, 0.000]
Clear onset range	[0.000, 0.000]
Has≥2 onset median	0.000
Clear onset median	0.000
Clear agreement crossing (25%)	0.000
Clear agreement crossing (50%)	0.000
Clear agreement crossing (75%)	0.000
Agreement width (75% − 25%)	0.000
Agreement width (75% − 50%)	0.000
Representative peaks (t_1, t_2)	(108, 321)
Peak separation $\mathrm{sep}_{\mathrm{peaks}}$	2.766
Peak balance ratio (min/max) / valley ratio	0.306 / 0.126
Directed peak ratio $R_{\mathrm{dir}} = \bar{f}(t_2)/\bar{f}(t_1)$ (diagnostic only)	0.306
Shortcut share around t_1 (± 20)	0.734
Shortcut share around valley (± 20)	0.596
Shortcut share around t_2 (± 20)	0.071
First $\leq 50\%$ shortcut-share time	155
Cumulative shortcut share at $t_{\max}$	0.121
Shortcut shift mass	0.060
Beta=0 relative-chain max diff	0.000
Has≥2 scaling slope / 95% CI	– / –
Clear scaling slope / 95% CI	– / –
Has≥2 / clear scaling R^2	– / –
$\varepsilon_{\mathrm{mass}}$	1.485e-16
$\sum_t f(t)$ at $t_{\max}$	0.99999994
$S(t_{\max})$	5.885e-08

The table now includes an explicit $\beta = 0$ cross-check against the reduced relative-coordinate chain; the max absolute discrepancy is at machine precision and validates the exact two-walker solver baseline.

To provide a concrete 1D ring shortcut instance in report form, we isolate the representative setup below: its parameter tuple, detector outputs, and explicit peak annotations are shown together. The first three prominent peaks (after smoothing) are listed for multi-peak visibility, while the bimodality decision still uses the detector pair (t_1, t_2).

Table 5: Representative 1D ring + shortcut instance (parameters and peak diagnostics).

Field	Value
N / q	101 / 0.70
(g_1, g_2)	(0.70, -0.40)
Starts (n_0, m_0)	(5, 55)
Shortcut (src $\rightarrow$ dst)	(5 $\rightarrow$ 70)
Representative β	0.20
Detector pair (t_1, t_2, t_v)	(108, 321, 278)
Prominent peaks (P_1, P_2, P_3)	(108, 192, 321)
Peak separation $\text{sep}_{\text{peaks}}$	2.766
Peak balance ratio (min/max) / valley ratio	0.306 / 0.126
Directed peak ratio $R_{\text{dir}} = \bar{f}(t_2)/\bar{f}(t_1)$ (diagnostic only)	0.306
Dominant splitting site / fraction	36 / 0.061

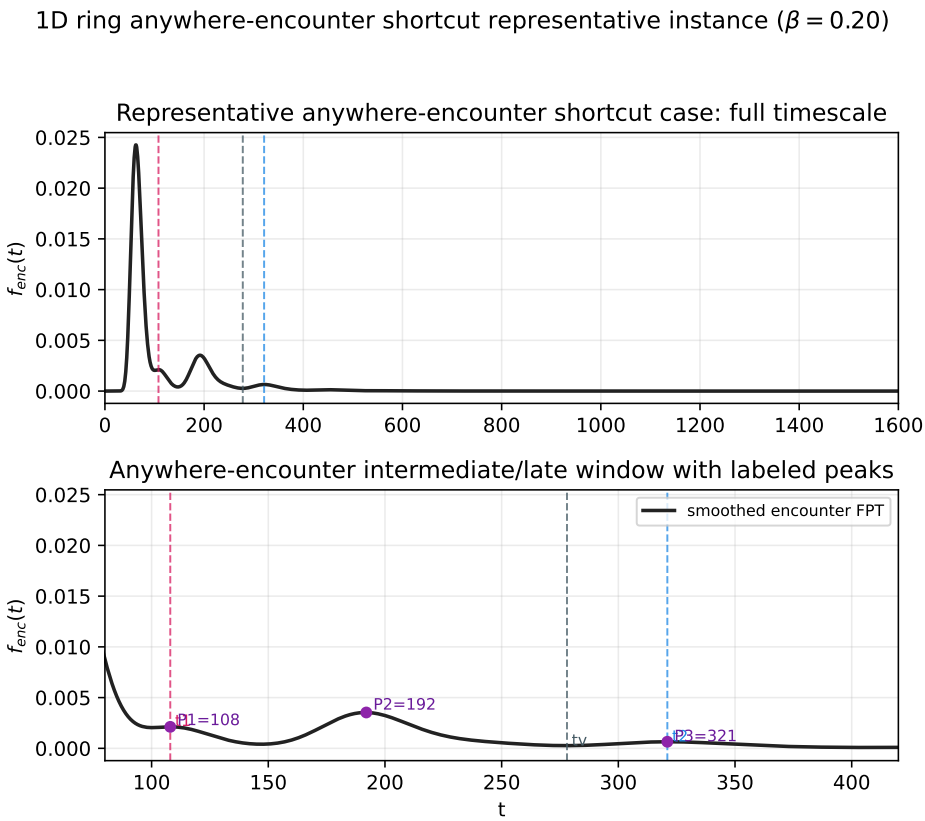

Figure 6: Representative anywhere-encounter case at $\beta = 0.20$ with labeled peak structure: detector pair (t_1, t_2) and prominent peaks P_1, P_2, P_3.

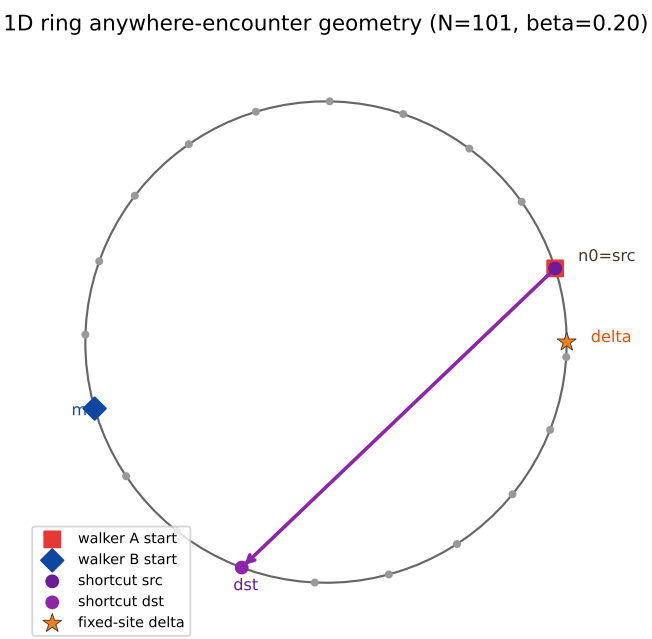

Figure 7: 1D ring anywhere-encounter geometry and shortcut placement; the fixed-site encounter point is marked as $\delta = 0$ (star).

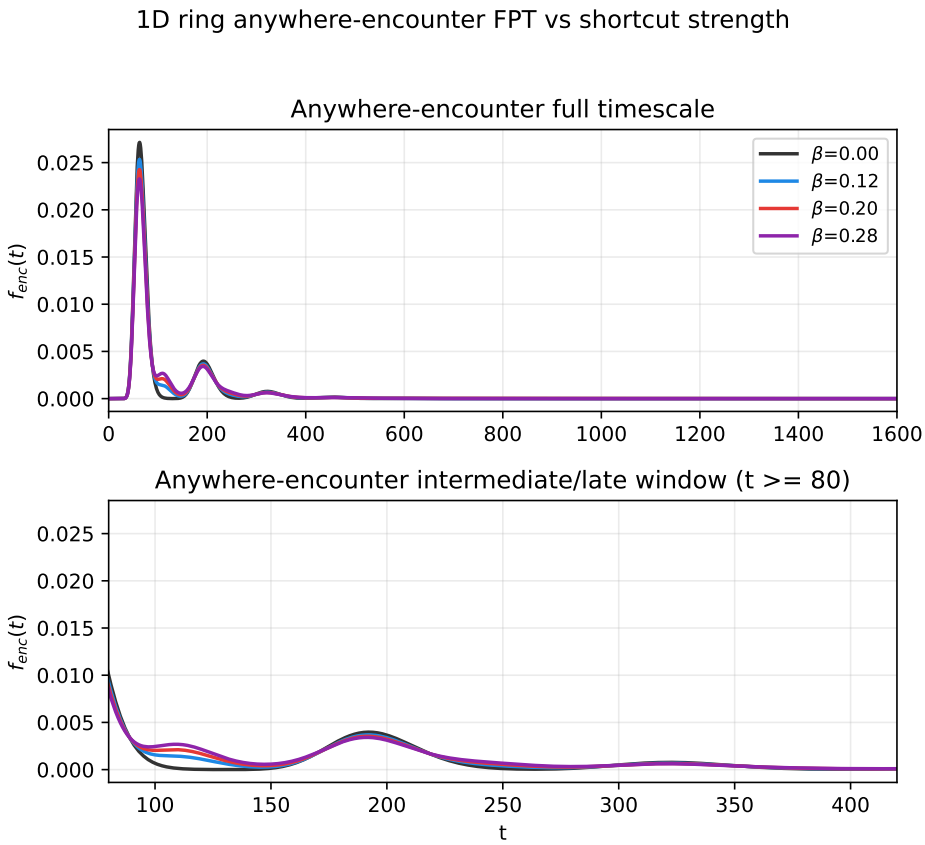

Figure 8: Anywhere-encounter FPT curves across shortcut strengths.

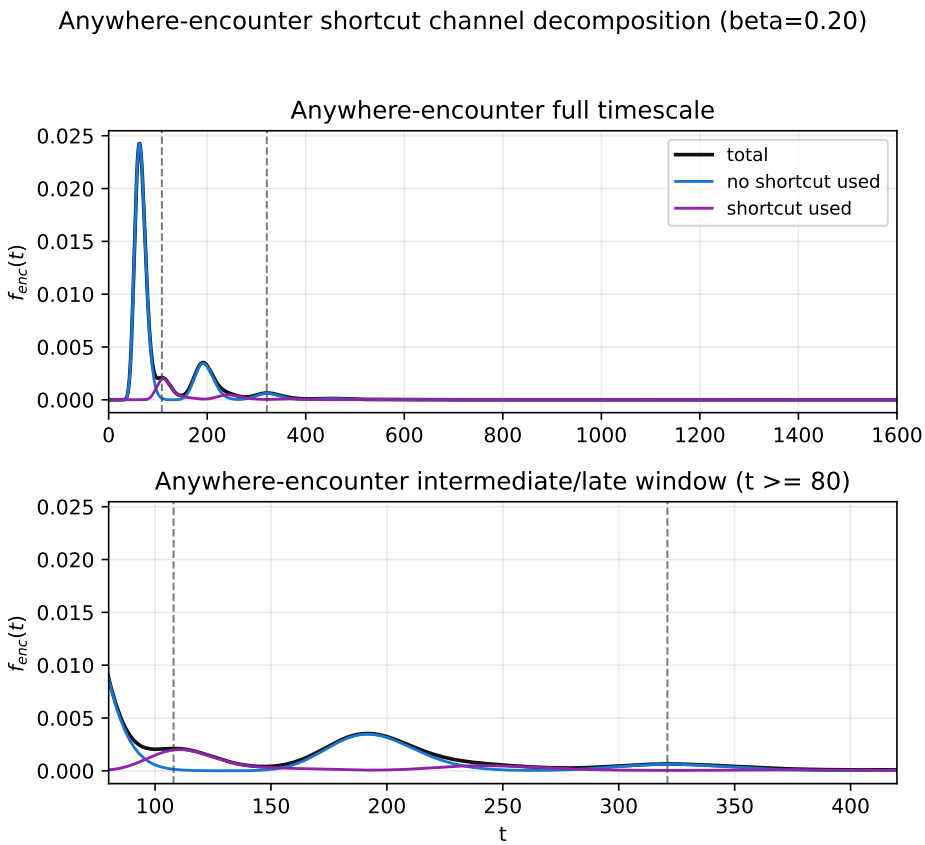

Figure 9: Anywhere-encounter shortcut-channel decomposition at $\beta = 0.20$: early lobe is shortcut-used dominated; late lobe is mostly no-shortcut trajectories.

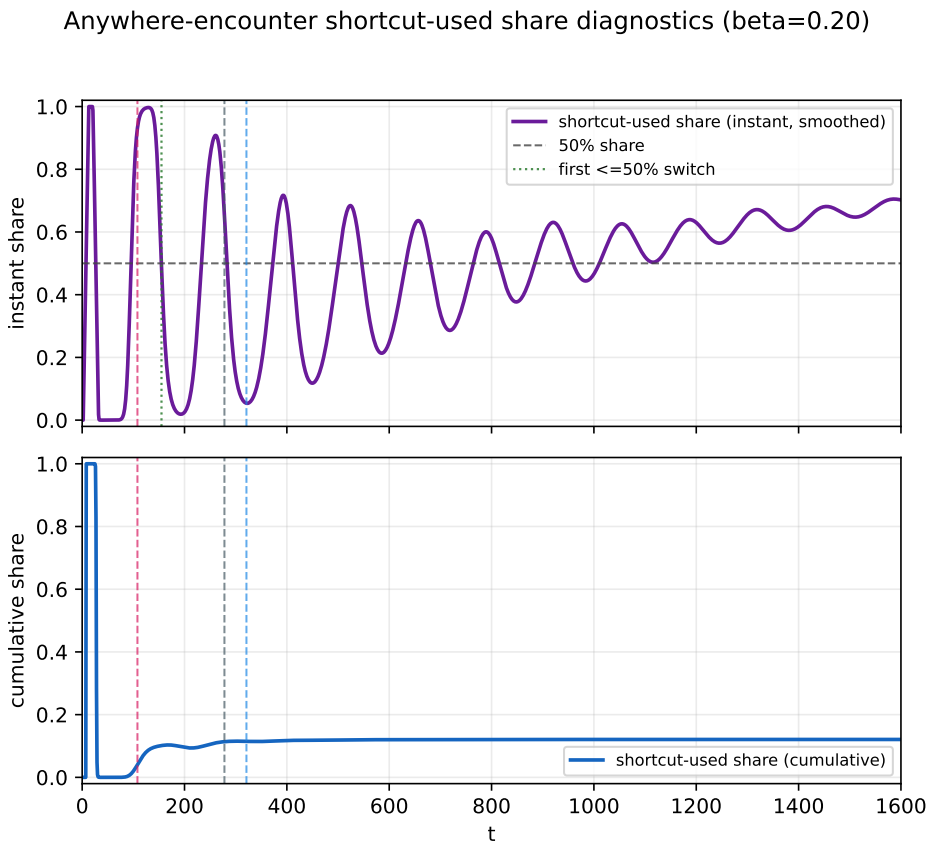

Figure 10: Anywhere-encounter shortcut-used share diagnostics at $\beta = 0.20$: instantaneous share (smoothed) and cumulative share versus time.

The share diagnostics make the channel handoff quantitative: shortcut-used trajectories dominate around the first peak and then give way to no-shortcut trajectories before the second peak, while the cumulative shortcut share remains bounded away from one.

To localize which timescales and which sites carry the full encounter signal, we record the diagonal first-encounter flux $\phi_t(n) = J_t(n, n)$ before absorption and define the site splitting probability

$$\pi_{\text{full}}(n) = \frac{\sum_t \phi_t(n)}{\sum_t f_{\text{enc}}(t)}. \tag{1}$$

This is the time-domain counterpart of Eq. (10) in [2]: the marker variable $\rho_{\boldsymbol{m}_j}$ selects the target contribution in the generating function, while evaluating the corresponding first-encounter flux at $z = 1$ yields the splitting probability over encounter sites. For each peak basin W_k selected by the timescale detector, we also define

$$\pi_k(n) = \frac{\sum_{t \in W_k} \phi_t(n)}{\sum_t f_{\text{enc}}(t)},$$

and analogously $\pi_{\text{yes}}(n), \pi_{\text{no}}(n)$ for shortcut-used / no-shortcut channels. The first three prominent basins are kept explicitly and the remaining mass is aggregated into 'other'.

Table 6: Representative peak-basin contributions at $\beta = 0.20$.

window	t_{left}	t_{peak}	t_{right}	frac. total	shortcut share	dom. site	dom. frac.
peak1	80	108	147	0.141	0.518	40	0.039
peak2	148	192	278	0.196	0.170	100	0.047
peak3	279	321	400	0.045	0.193	63	0.035
other	–	–	–	0.617	0.009	36	0.092

Table 7: Representative site splitting probabilities for the dominant encounter sites.

site	p_{full}	p_{peak1}	p_{peak2}	p_{peak3}	p_{other}	p_{yes}
36	0.061	0.004	0.000	0.000	0.057	0.001
37	0.060	0.005	0.000	0.000	0.056	0.001
35	0.058	0.004	0.000	0.000	0.054	0.001
38	0.057	0.005	0.000	0.000	0.052	0.000
34	0.053	0.003	0.000	0.000	0.049	0.001
39	0.052	0.005	0.000	0.000	0.046	0.000
33	0.045	0.003	0.000	0.000	0.042	0.001
40	0.044	0.005	0.000	0.000	0.039	0.000
41	0.037	0.005	0.000	0.000	0.031	0.000
32	0.037	0.003	0.000	0.000	0.034	0.002

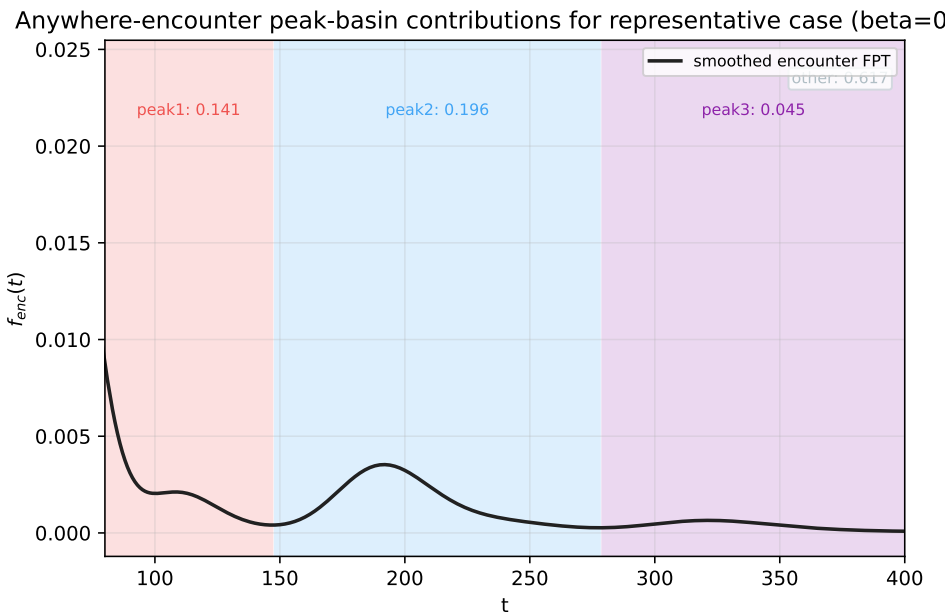

Figure 11: Representative anywhere-encounter peak-basin partition: the first three prominent basins are shown explicitly, while the residual encounter mass is reported as 'other'.

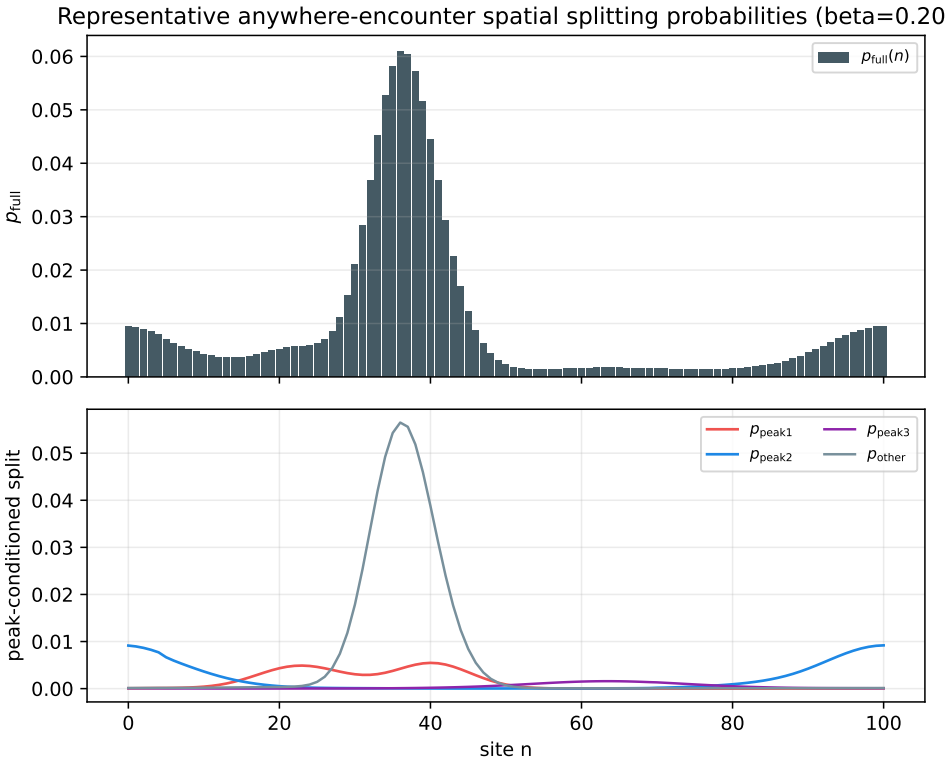

Figure 12: Representative anywhere-encounter site splitting profile from Eq. (1), resolved into peak-conditioned contributions.

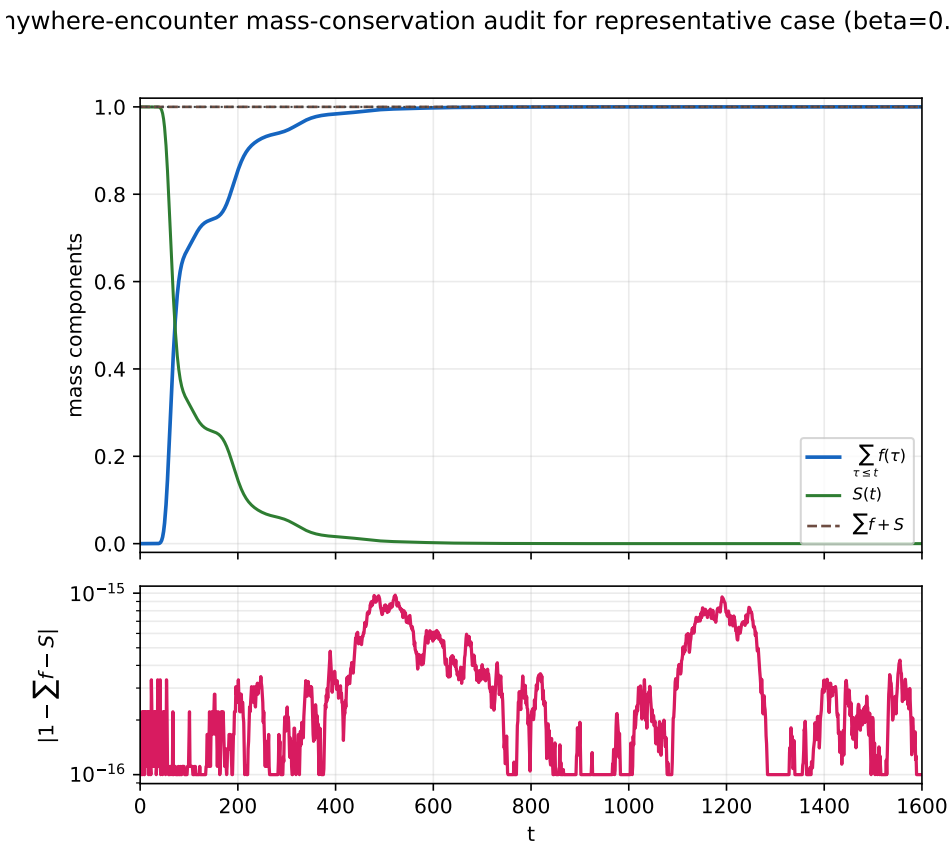

Figure 13: Anywhere-encounter mass-conservation audit for the representative case: cumulative encounter mass, survival tail, and residual $|1 - \sum_{\tau \le t} f(\tau) - S(t)|$.

The residual curve stays near machine precision and confirms that the late-time shoulder is a model effect rather than truncation leakage.

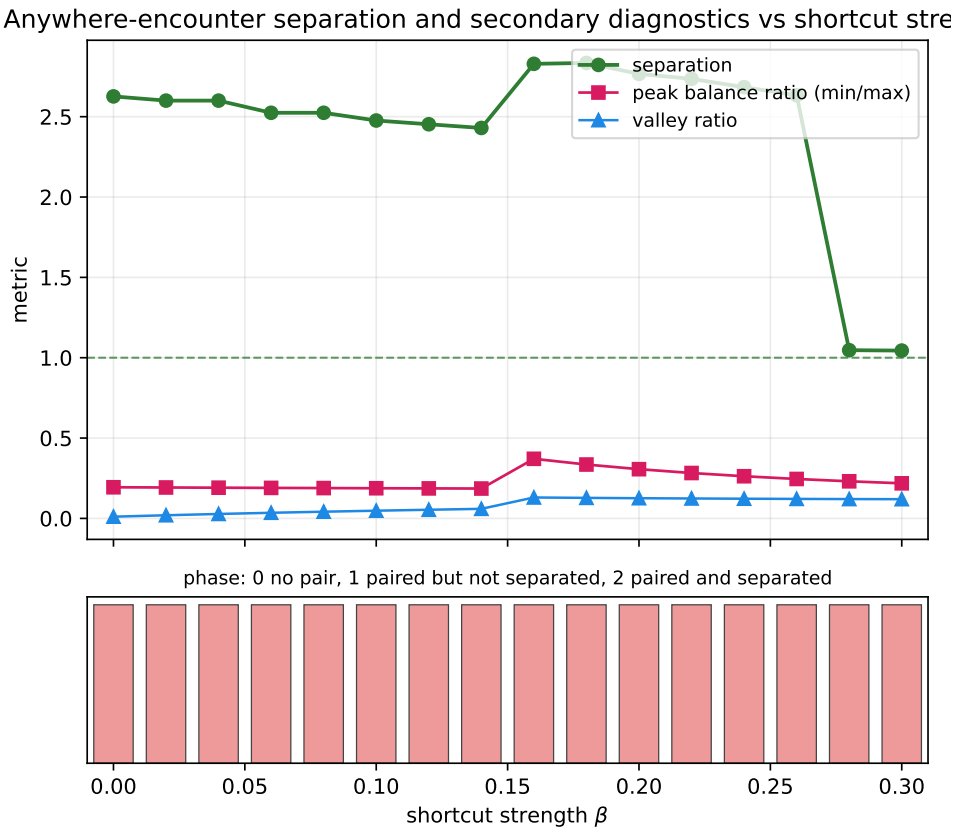

Figure 14: Anywhere-encounter peak-balance (min/max) / valley metrics and phase labels versus β (timescale detector, primary).

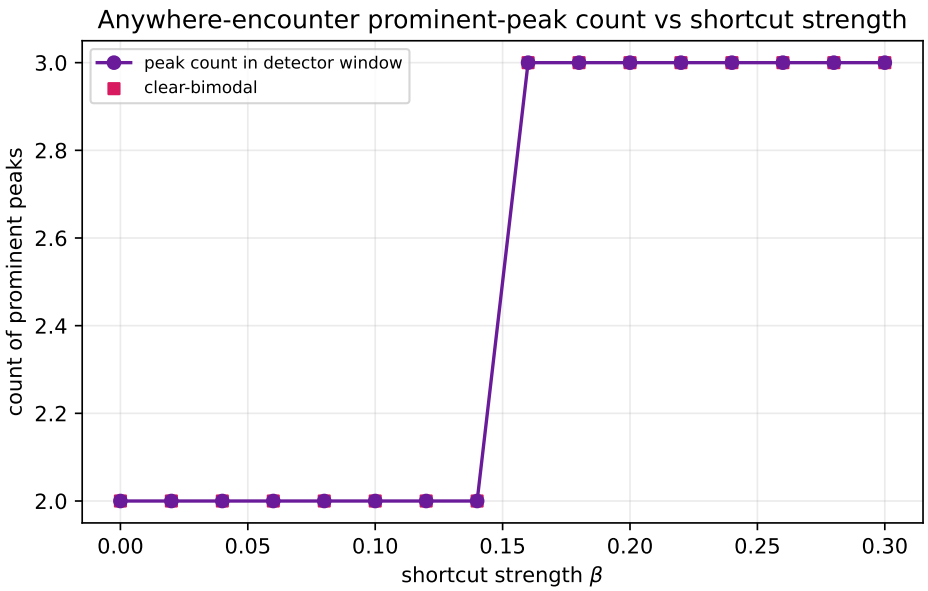

Figure 15: Anywhere-encounter number of prominent peaks in the detector window versus β (timescale detector view): the regime shifts from mostly 2 peaks to 3 peaks around onset.

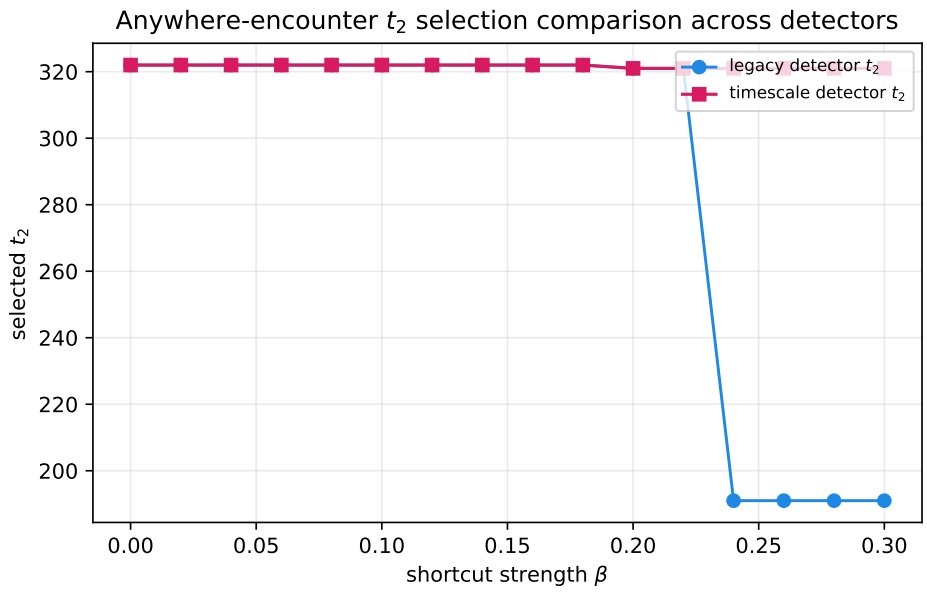

Figure 16: Anywhere-encounter t_2 selection comparison: legacy second-peak rule can switch from late to middle peak near $\beta \approx 0.24$, while the timescale detector remains stable on the late scale.

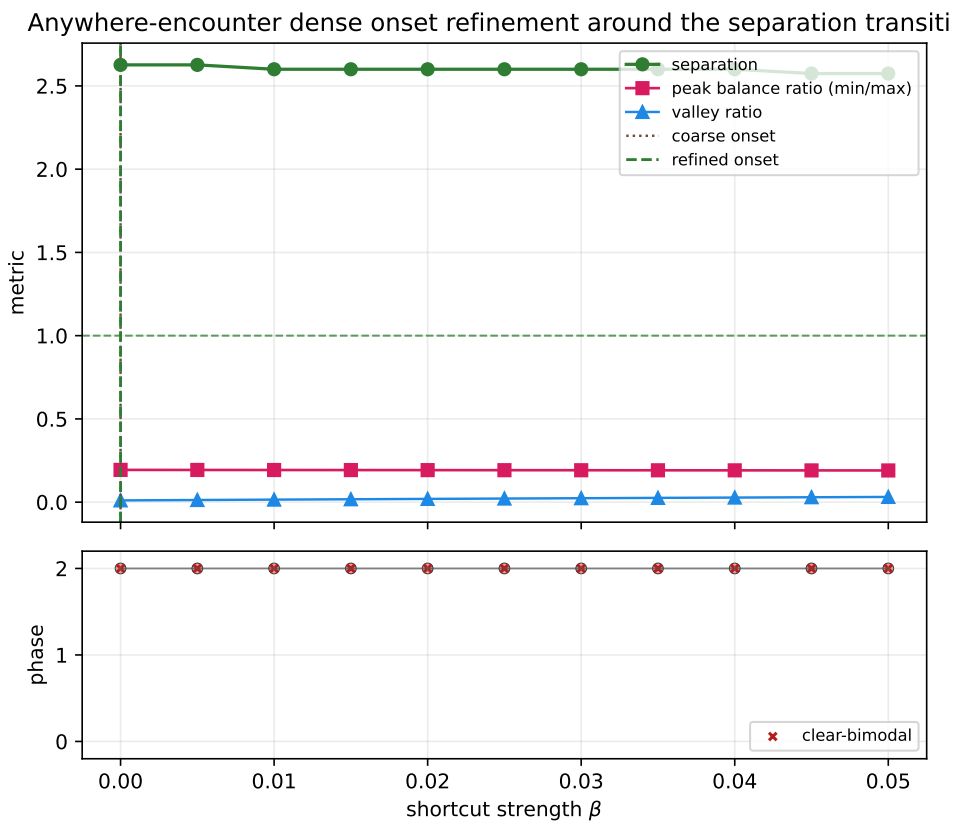

Figure 17: Anywhere-encounter dense refinement around onset: metric curves and phase labels on the local β window.

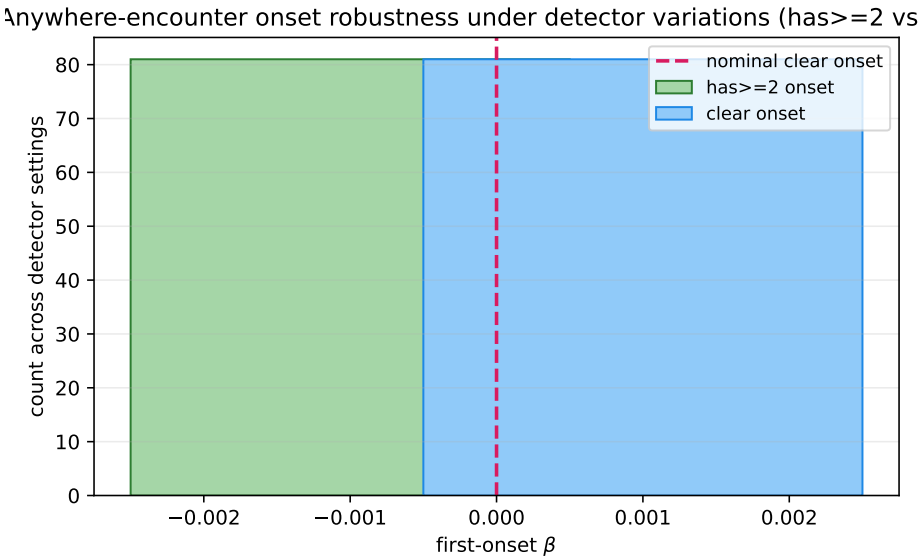

Figure 18: Anywhere-encounter distribution of 'has≥2' onset and clear onset under detector-parameter variations; dashed line marks the nominal clear refined onset.

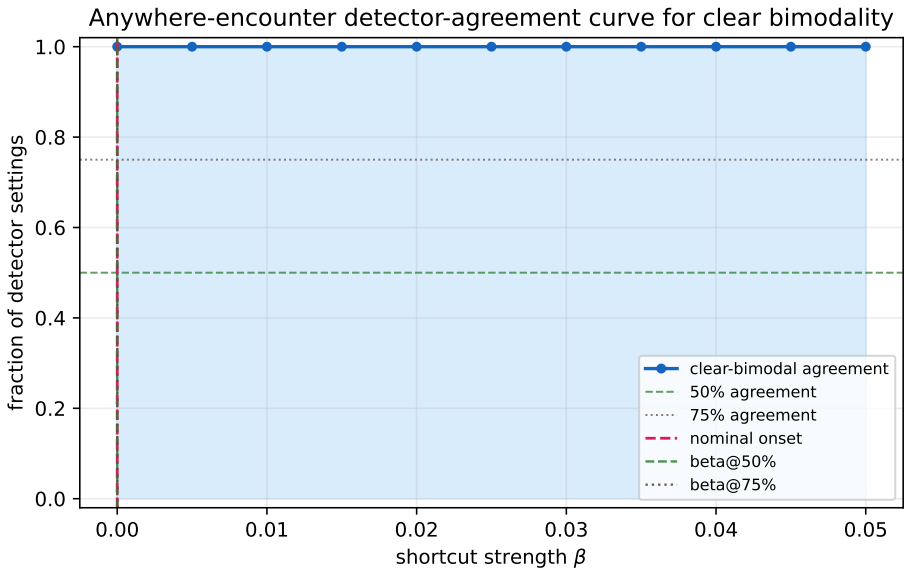

Figure 19: Anywhere-encounter detector-agreement curve across refined β: crossings at 25%, 50%, and 75% define onset location and transition width.

Ring-size robustness conclusions are also auto-generated to avoid manual drift: To avoid over-reliance on the single baseline size $N = 101$, we additionally scan $N \in \{61, 81, 101, 121, 141, 161, 181, 201\}$ with geometry-scaled $(n_0, m_0, \text{src}, \text{dst})$. For each N, we recompute has≥2 onset, clear onset, and clear-bimodal coverage on the nominal $\beta \in [0, 0.30]$ grid using the same timescale selector as the main scan (first peak + score-selected t_2), then compare representative peak/valley ratios at $\beta = 0.20$; if no clear onset appears, we extend once to $\beta \leq 0.50$ (step 0.02). Current run summary: 8 size points total, 8 with nominal-window has≥2 onset, 8 with nominal-window clear onset, 0 recovered by extension, 0 still unresolved. Extension details: No extension-only onset recovery is needed in this run. Nominal-window conclusion: $N \in \{61, 81, 101, 121, 141, 161, 181, 201\}$ all

show clear onset inside the nominal window. The resulting trend is summarized in Fig. 20 and Table 8.

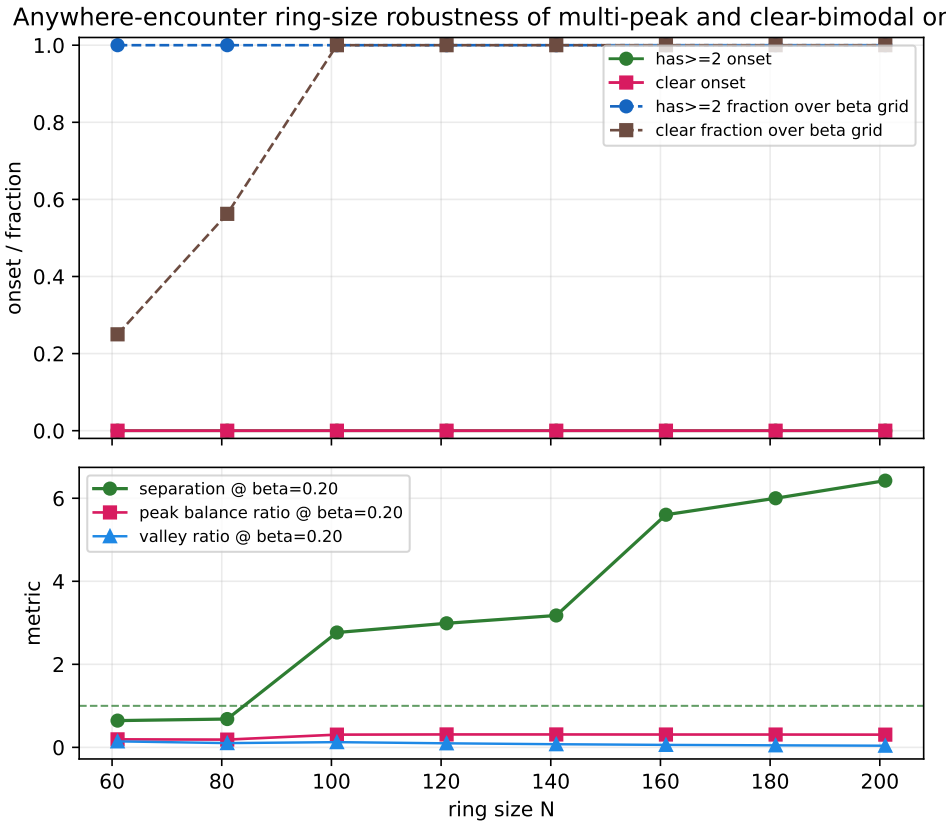

Figure 20: Anywhere-encounter ring-size robustness scan: 'has≥2' onset, clear onset, and clear-fraction trends across N, plus representative ratio diagnostics at $\beta = 0.20$ (star marker = clear onset recovered by extension beyond $\beta = 0.30$).

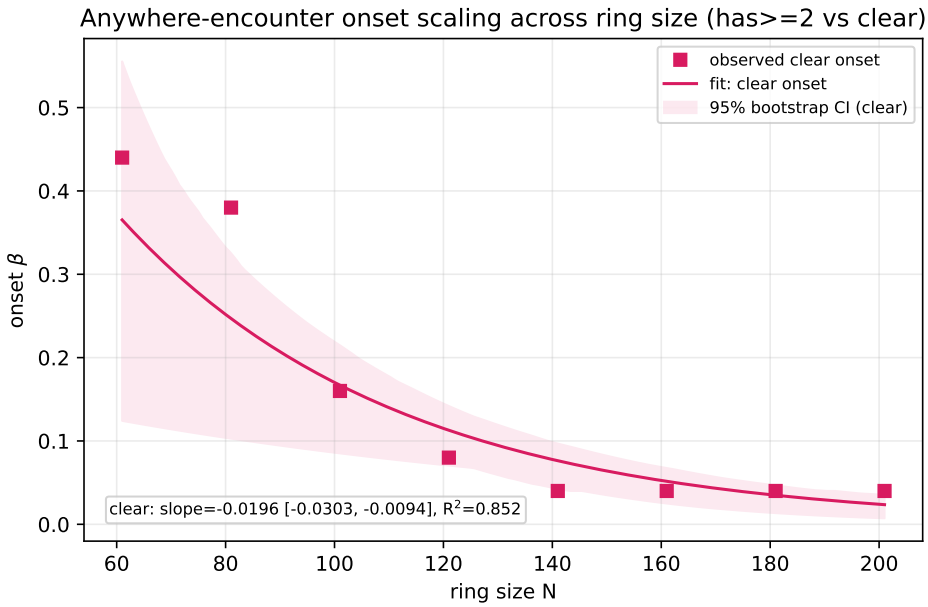

Figure 21: Anywhere-encounter onset scaling against ring size: separate fits for 'has≥2' onset and clear onset in $(N, \log \beta_{\text{onset}})$ coordinates, with bootstrap confidence intervals when resolvable.

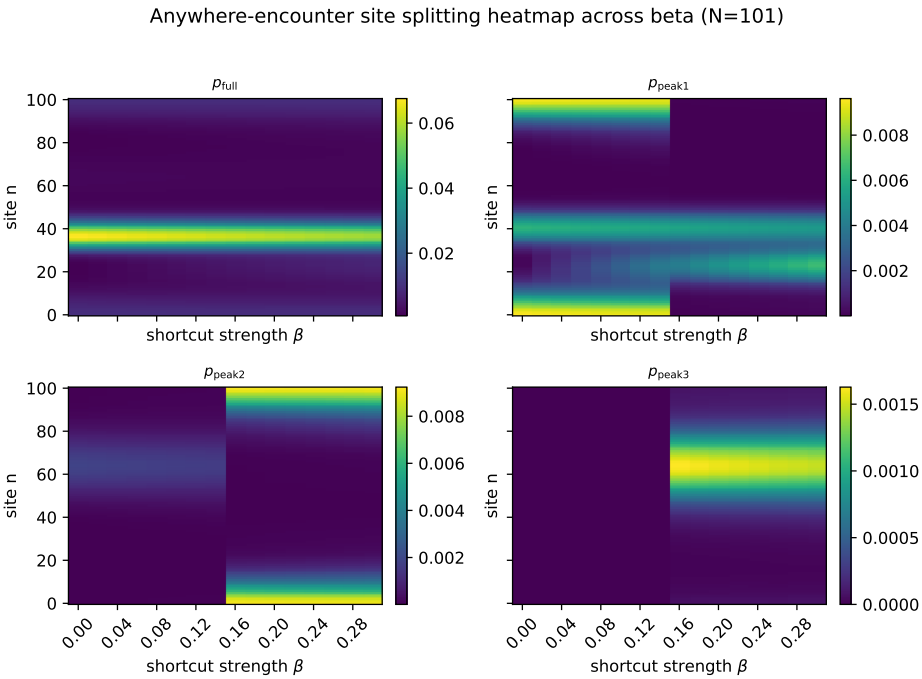

Figure 22: Anywhere-encounter site splitting heatmap versus shortcut strength at fixed $N = 101$.

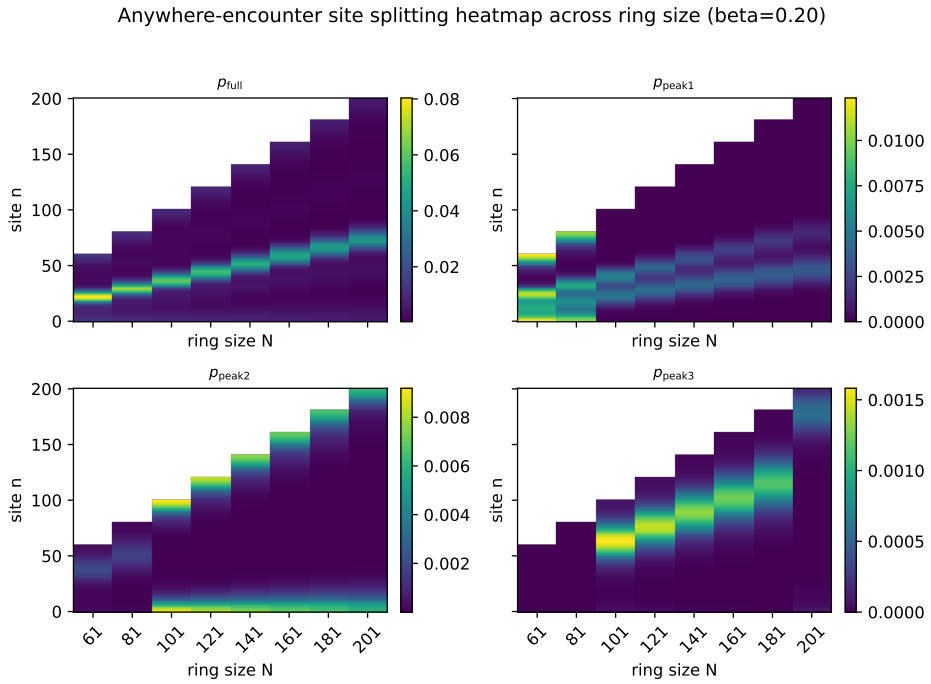

Figure 23: Anywhere-encounter site splitting heatmap versus ring size at fixed $\beta = 0.20$.

To make extension usage explicit rather than implicit, Fig. 24 separates nominal-window detections from extension-recovered and unresolved cases, and overlays the actual per-N search cap $\beta_{\max}(N)$. This avoids interpreting "no onset in $[0, 0.30]$" as a hard physical absence when it can be a window effect. Table 8 now also exposes the per-row onset source tag ('main/extended/none') and the corresponding $\beta_{\max}$, so extension recovery can be audited without cross-checking raw CSV files.

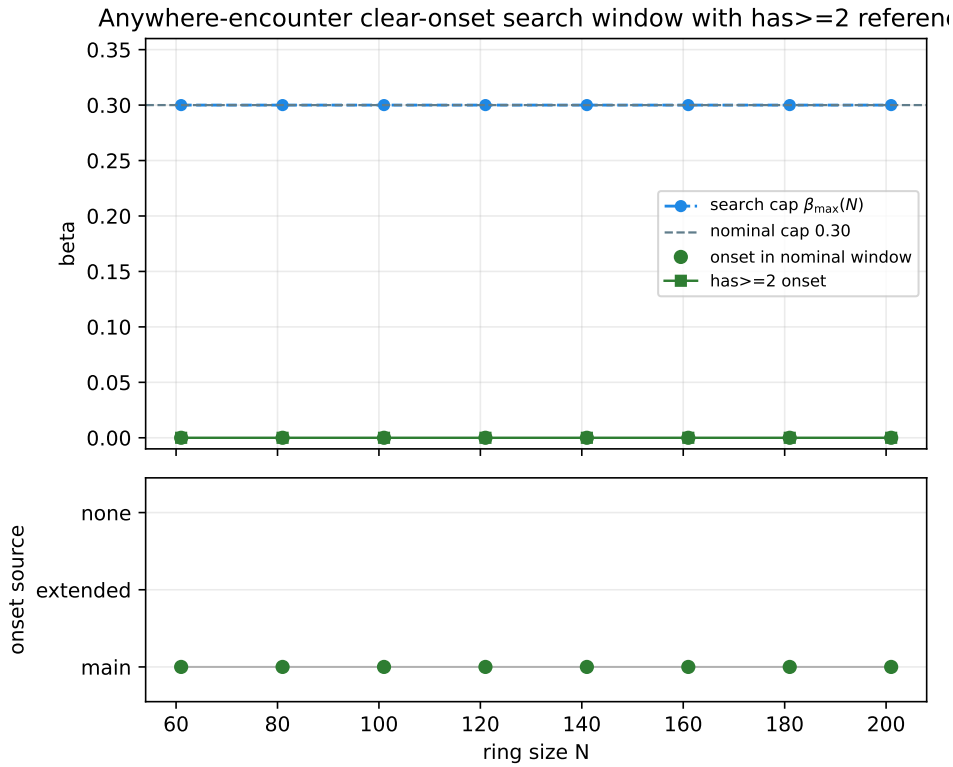

Figure 24: Anywhere-encounter onset-source diagnostics across ring sizes: top panel shows resolved onset versus per-N search cap and nominal cap 0.30; bottom panel labels whether each onset is found in the nominal window, recovered by extension, or still unresolved.

Table 8: Ring-size onset robustness summary with explicit onset source and search cap columns ($\beta^{\dagger}$ = resolved onset; "*" marks extended onset; "$> x$" means no clear onset up to search cap x).

N	has≥2 onset	clear onset	clear source	$\beta_{\max}$	has≥2 frac	clear frac	phase@0.20	sep@0.20	pe
61	0.000	0.000	`main`	0.300	1.000	0.250	1	0.644	
81	0.000	0.000	`main`	0.300	1.000	0.562	1	0.682	
101	0.000	0.000	`main`	0.300	1.000	1.000	2	2.766	
121	0.000	0.000	`main`	0.300	1.000	1.000	2	2.988	
141	0.000	0.000	`main`	0.300	1.000	1.000	2	3.176	
161	0.000	0.000	`main`	0.300	1.000	1.000	2	5.603	
181	0.000	0.000	`main`	0.300	1.000	1.000	2	6.000	
201	0.000	0.000	`main`	0.300	1.000	1.000	2	6.424	

Table 9: Shortcut scan metrics and phase labels.

β	has≥2	phase	t_1	t_2	sep	peak balance ratio (min/max)	valley ratio	separated
0.00	1	2	192	322	2.626	0.194	0.011	1
0.02	1	2	192	322	2.600	0.192	0.020	1
0.04	1	2	192	322	2.600	0.191	0.028	1
0.06	1	2	192	322	2.524	0.190	0.035	1
0.08	1	2	192	322	2.524	0.189	0.042	1
0.10	1	2	192	322	2.476	0.188	0.048	1
0.12	1	2	192	322	2.453	0.187	0.054	1
0.14	1	2	192	322	2.430	0.186	0.060	1
0.16	1	2	107	322	2.829	0.371	0.130	1
0.18	1	2	108	322	2.834	0.335	0.128	1
0.20	1	2	108	321	2.766	0.306	0.126	1
0.22	1	2	109	321	2.735	0.283	0.124	1
0.24	1	2	109	321	2.684	0.262	0.122	1
0.26	1	2	109	321	2.634	0.245	0.121	1
0.28	1	2	109	321	1.047	0.231	0.120	1
0.30	1	2	109	321	1.044	0.218	0.120	1

7 Pitfalls and Debug Checklist

1. **Contour crossing poles:** abort if $r_{\text{in}} \geq r_{\text{out}}$.
2. **Near-singular amplification:** monitor $\min_{k_1,k_2} |1 - z\alpha_{k_1}\alpha_{k_2}|$.
3. **False early micro-peaks:** keep fixed smoothing and t_{ignore}.
4. **Truncation bias:** always report $\sum_t f(t)$ and survival tail.

8 Positioning Against Luca (Bristol) Literature

This report places encounter dynamics into the same finite-domain random-walk first-passage framework used in Luca Giuggioli's line of work:

- The fixed-site renewal ratio
$$\widetilde{f}_\delta(z) = \frac{\widetilde{Q}_{n_0,m_0}(z)}{\widetilde{Q}_{\delta,\delta}(z)}$$
is consistent with multi-target/heterogeneous first-passage decompositions[3].
- The A1/A8 generating-function product route is compatible with defect/heterogeneity-oriented analytical strategies in related first-passage studies[1].

So the encounter report is aligned with literature at the level of definitions, measurable diagnostics, and analytic-numeric closure, while adding a shortcut-channel evidence chain specific to this project.

9 Reproducibility

```
python3 scripts/reportctl.py run --report ring_two_walker_encounter_shortcut -- python3 code
python3 scripts/reportctl.py build --report ring_two_walker_encounter_shortcut --lang cn
python3 scripts/reportctl.py build --report ring_two_walker_encounter_shortcut --lang en
```

References

[1] S. Sarvaharman and L. Giuggioli, *First-passage-time statistics of a random walker under biased quenched disorder*, Phys. Rev. E **102**, 062124 (2020). DOI: 10.1103/PhysRevE.102.062124.

[2] L. Giuggioli and S. Sarvaharman, *Spatio-temporal dynamics of random transmission events: from information sharing to epidemic spread*, J. Phys. A: Math. Theor. **55**, 375005 (2022). DOI: 10.1088/1751-8121/ac8587.

[3] L. Giuggioli, S. Sarvaharman, D. Das, D. Marris, and T. Kay, *Multi-target search in bounded and heterogeneous environments: a lattice random walk perspective*, arXiv:2311.00464v2 [cond-mat.stat-mech] (2024).